TEAM 2

SCOTBANK

CLOUD DEPLOYMENT STRATEGY

CS217



WHERE WE SHOULD DEPLOY

We should deploy our Portfolio management system on a public cloud platform, such as AWS, Azure or Google Cloud. These platforms use a multi-tenant model and so the costs will be lower than other options while still allowing us to have access to large-scale infrastructure. We should also consider using a cloud region close to our target users in Scotland to reduce latency, we should be using UK or EU data centers.

WHY CHOOSE THIS OPTION FOR DEPLOYMENT?

Benefits are:

- **Scalability**: cloud platforms support rapid elasticity, meaning the application can automatically scale up when many users are trading investments and scale down during quieter periods. This keeps costs predictable.
- **Cost Efficiency**: Public cloud follows a pay-as-you-go model, so we would only pay for the compute, storage, and networking that we actually use. This is especially helpful for smaller platforms like us.
- **Reliability and High Availability**: Because public cloud providers operate with multiple Availability Zones, we can deploy our application across them so that if one fails, the others take over. Load balancers and auto-scaling also improve reliability.
- **Managed Services:** The cloud provider will handle hardware, patching, networking, and security updates, which is ideal for a new and smaller platform since we likely do not have the manpower to maintain complex infrastructure.
- **Security Features:** Although of course users are still responsible for some security tasks, public clouds offer identity management, encryption tools, and automatic security patching, which helps meet basic financial-industry expectations.

HOW WE SHOULD DEPLOY

We should deploy our application using the **PaaS model**, with PaaS the cloud provider will manage: the server hardware, the operating system, security updates, networking and storage and middleware.

With this handled by a 3rd party we have more resources to put into writing, deploying and improving our application.

DEVELOPMENT WORKFLOW OVERVIEW

1. Build our application

   Develop our Portfolio Management System
2. Package the app

   We can package our application in containers so we can host microservices
3. Deploy to a PaaS service

   By using a PaaS service such as Azure or AWS scaling and deployment will be handled for us
4. Configure an Application Load Balancer

   This is so that traffic is split across multiple instances and points of failure get removed
5. Enable Auto-Scaling

   We can set up systems so that more instances are added if CPU usage is too high or to remove instances if we are getting low traffic
6. Set-up Multiple Availability Zones

   By setting up multiple availability zones we can protect against outages, this will give us redundancy and failover.

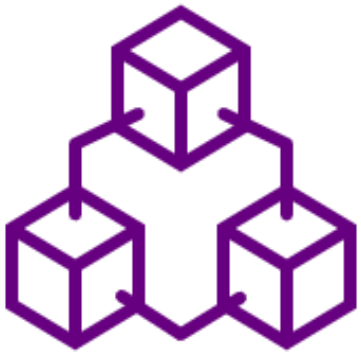
CLOUD SERVICES WE WILL USE

By deploying to cloud we can take advantage of many cloud services



AUTO-SCALING GROUPS

Automatically increase or decrease application instances based on demand. Improves responsiveness during high traffic (e.g., market opening hours).



APPLICATION LOAD BALANCER

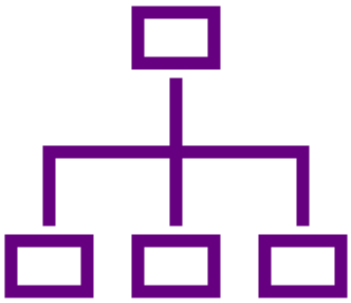
Balances user traffic across multiple servers and performs health checks. Reduces latency and avoids overloading a single instance.



CONTAINERS + MICROSERVICES

Allow each part of our application to scale independently. Containers provide isolated, consistent environments for each service. This improves performance because only high-demand parts need extra resources







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SERVERLESS FUNCTIONS FOR MARKET DATA

We can serverless functions to fetch real-time stock data. They will run only when needed, react quickly and avoid long running background services which improves cloud elasticity.

MANAGED DATABASES WITH REPLICATION

They allow us to automatically replicate data across regions and availability zones. This will improve read/write performance, reliability and disaster recovery responsiveness.

MULTIREGION DEPLOYMENT

Allow requests to be routed to the nearest or healthiest region using Route 53 latency or geography routing. This will improve performance and reliability globally.

SUMMARY

We start by selecting a public cloud environment, which will give us cost-effective infrastructure, strong security controls and the ability to operate across multiple Availability zones for improved reliability.

The system will be built and deployed using a PaaS (Platform as a Service) approach, this will simplify operations by allowing the platform to handle core infrastructure and lets us keep our manpower focused on the application.

Once deployed our application will run behind an application load balancer, ensuring traffic is evenly distributed and unhealthy instances are removed automatically. Auto-scaling will be used to adjust our compute capacity in response to traffic, meaning we can maintain our performance during peak hours without unnecessary cost.

Our design uses containers and microservices so that our individual components like portfolio calculations or market data retrieval can scale independently as the usage grows. A serverless function will be used to handle external market data efficiently, while a managed database will provide secure storage and redundancy across multiple availability zones. For risk reduction we can also support multi-region failure using DNS based routing, this will mean our system can stay available even when there is regional disruption.